
Software Requirements Specification

for

Online Food Delivery Management System

Version 1.0 approved

Prepared by Abdul Haseeb

Karachi Institute of Economics & Technology (KIET)

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Revision History

Name	Date	Reason For Changes	Version
Abdul Haseeb	5/5/2026	Initial Draft	1.0
Abdul Haseeb	11/5/2026	Updated Usecase and Activity Diagram	1.1
Abdul Haseeb	16/5/2026	Use case realization and updated sequence diagram	2.0
Abdul Haseeb	21/5/2026	Updated sequence diagram, Class diagram	2.1
Abdul Haseeb	01/6/2026	Final Changes	3.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification document is to provide a detailed and complete overview of the Online Food Delivery Management System. This document specifies both the functional and non-functional requirements of the system based directly on the established use case design.

The scope of this SRS covers the entire web application system, which is designed to streamline the food ordering and delivery process. It details the complete interactions and workflows of three primary user roles: the Customer, the Restaurant Manager, and the Delivery Rider.

1.2 Document Conventions

The following conventions are used in this document:

- Headings are written using Times font.
- Body text is written using Arial font.
- Functional requirements are grouped under System Features.
- Requirements are categorized based on system functionality and priority levels.

These conventions are strictly followed to maintain structured and easy to navigate documents.

1.3 Intended Audience and Reading Suggestions

This document is designed for different stakeholders involved in the design, development, and deployment of the Online Food Delivery Management System. The intended audience includes:

- **Project Managers:** To review and verify that all system scopes align perfectly with the structural design and use case boundaries.
- **Software Developers:** To understand the exact functional flow, include/extend relationships, and logical validations required for implementation.
- **Quality Assurance:** To draft precise test cases and user acceptance tests based strictly on the specified use case boundaries.

1.4 Product Scope

The software being specified is the Online Food Delivery Management System. It is a self-contained digital platform designed to bridge the gap between hungry customers, restaurant management, and delivery personnel.

System Objectives and Goals:

For Customers: To provide a seamless platform where they can register/login, browse menus, apply promotional codes, place orders securely, track their delivery rider on a live map, and give feedback on their experience.

For Restaurant Managers: To fully digitize backend operations, allowing them to manage menu items, update prices, process or reject incoming orders, and dynamically assign riders to confirmed orders.

For Delivery Riders: To efficiently manage daily tasks by viewing customer locations, updating delivery statuses in real-time, and using built-in Google Maps navigation.

1.5 References

The development of this Software Requirements Specification document is based on the following foundational documents, technical standards, and modeling frameworks:

System Use Case Model Diagram: The primary architectural diagram provided in the project design phase, detailing all links between the Customer, Restaurant Manager, and Delivery Rider.

UML Specification: Unified Modeling Language standard guide used to define the boundaries, actors, and specific structural relationships within the system.

IEEE Std: IEEE Recommended Practice for Software Requirements Specifications. Used as the underlying structural standard for drafting the sections of this document.

2. Overall Description

2.1 Product Perspective

The Online Food Delivery Management System is a new, standalone software system. It is not a part of any old system, nor does it depend on any existing software family. Instead, it is built from scratch to replace the old manual way of taking food orders over phone calls or paper receipts.

The system connects three different users together: the Customer, the Restaurant Manager, and the Delivery Rider. Everything works together under one centralized system based exactly on our use case diagram. The only major external service this software uses is the Google Maps

API. This external tool is necessary because it allows the Customer to *View live map* and helps the Delivery Rider to *Use Google Maps* to find the customer's address easily.

2.2 Product Functions

The major functions of the system are grouped by the core operational responsibilities of the three primary actors as illustrated in the system design.

1. Customer Functions

Account Management: Allows new users to *Signup* and existing users to *Login*, with both processes including a *View details* step for profile or validation purposes.

Menu Interaction: Enables customers to *Browse Menu* and view specific food items.

Order Placement: Allows users to *Place Order*. This includes an extended feature to *Apply promo code* if available. The process also includes a mandatory *Make payment* step, which can trigger a *Reject Order* action if payment fails.

Order Fulfillment & Feedback: Enables users to *Track Order* and finally *Rate or Review* the service after completion.

2. Restaurant Manager Functions

Menu Management: Provides tools to *Manage Menu*, which includes sub-actions to *add or remove item* and *Update Prices*.

Order Management: Allows the manager to *Process order*. This involves an included step to *Confirm availability* of items and the authority to *Reject Order* under custom exceptions.

Rider Assignment: Enables the manager to manually or dynamically *Assign Order* to available delivery personnel once the order is confirmed and prepared.

3. Delivery Rider Functions

Status Management: Allows the rider to *Update Status* of the order at various checkpoints, which includes marking the order as *Marked picked up* or *Mark marked delivered*.

Navigation & Fulfillment: Enables the rider to *View Location* of the customer, which includes an automatic dependency to *Use Google Maps* for precise turn-by-turn routing.

2.3 User Classes and Characteristics

The system has three main classes of users. Each class has different access levels, features, and responsibilities within the application:

1) Customer:

Functions Used: Login/Signup, Browse Menu, Place Order, Apply Promo Code, Track Order, and Rate or Review.

Characteristics: These are general public users with varying levels of technical skills. The system UI must be extremely simple and intuitive for them. They only have privileges to manage their own profiles and view or place their own orders.

2) Restaurant Manager:

Functions Used: Manage Menu add/remove items, Update Prices, Process Order Confirm availability, Reject Order, and Assign Order.

Characteristics: These users are internal staff members who have moderate technical knowledge. They have higher security access because they control the restaurant's menu prices and handle order processing and rider assignments.

3) Delivery Rider:

Functions Used: Update Status Marked picked up, Mark delivered and View Location.

Characteristics: These are field workers who use the system entirely on mobile devices while on the road. They require a mobile-friendly interface with large buttons and direct integration with navigation tools. They have specific access to view only the delivery details and locations assigned to them.

Use Case Realization:

a) Login / Signup

Boundary (Authentication Form):

Take user credentials (username/password) and displays error messages if login fails.

Control (Authentication Manager):

Verifies credentials against the database and manages session.

Entity (User Account):

Stores sensitive data including username and passwords.

b) Place Order:

Boundary (Order Form):

Handles user inputs for address, payment selection, and displays the generated Order ID.

Control (Order Manager):

Manages the validation details, calculating totals, and coordinating with the restaurant for availability.

Entity (Order & Menu):

Stores order details (items, price, status) and provides menu data for selection.

c) Process Order:

Boundary (Manager Options):

Provides the interface for the Restaurant Manager to view incoming orders and confirm item availability and then assign order to rider.

Control (Workflow Manager):

Show the receive Orders.

Entity (Order & Inventory):

Updates the order state and checks stock for the requested items.

d) Track Order:

Boundary (Tracking Option):

Displays the live map and real-time status updates to the customer.

Control (Location control):

Fetches GPS coordinates from the Delivery Rider and pushes updates to the Customer's view.

Entity (Delivery):

Maintains the relationship between a specific order and the rider's current data.

e) Manage Menu:

Boundary (Menu Management Option):

Allows the Manager to add/remove items and update prices.

Control (Menu Controller):

Validates price changes and updates the system-wide catalog.

Entity (Menu):

The persistent storage containing item descriptions, categories, and pricing history.

f) Update Deliver Status:

Boundary (Boundary App Options):

The interface where the rider marks an order as Picked Up or Delivered.

Control (Status Controller):

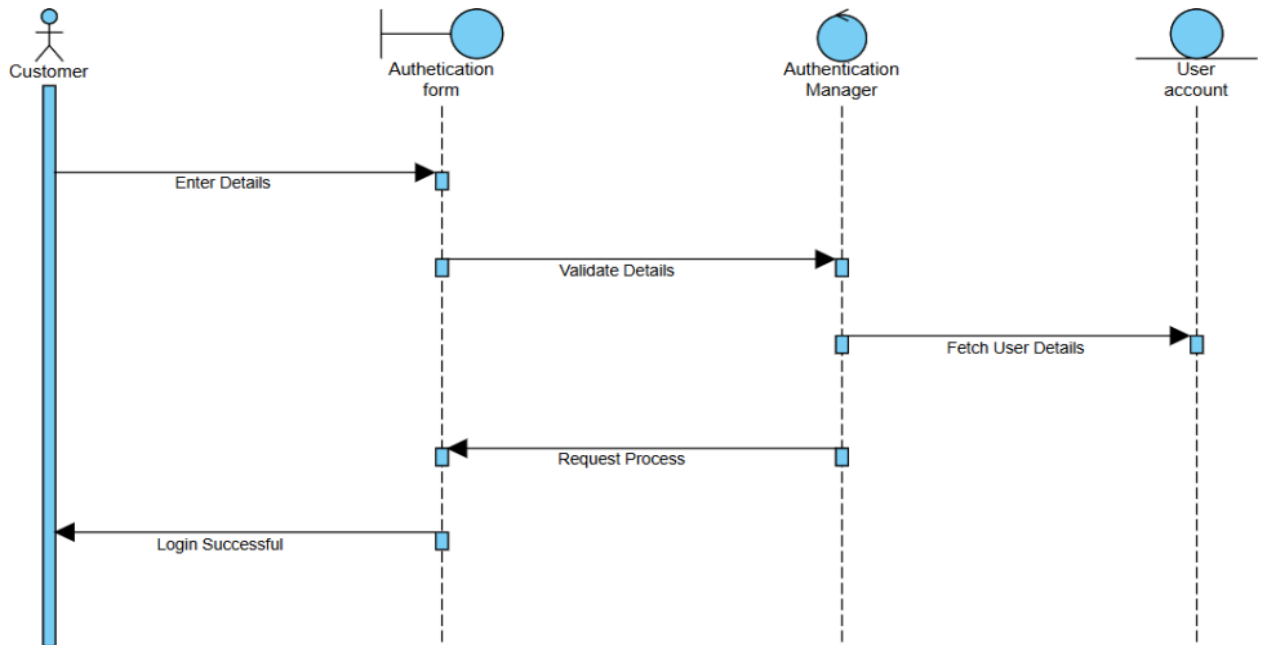
Update rider's status.

Entity (Order & Rider):

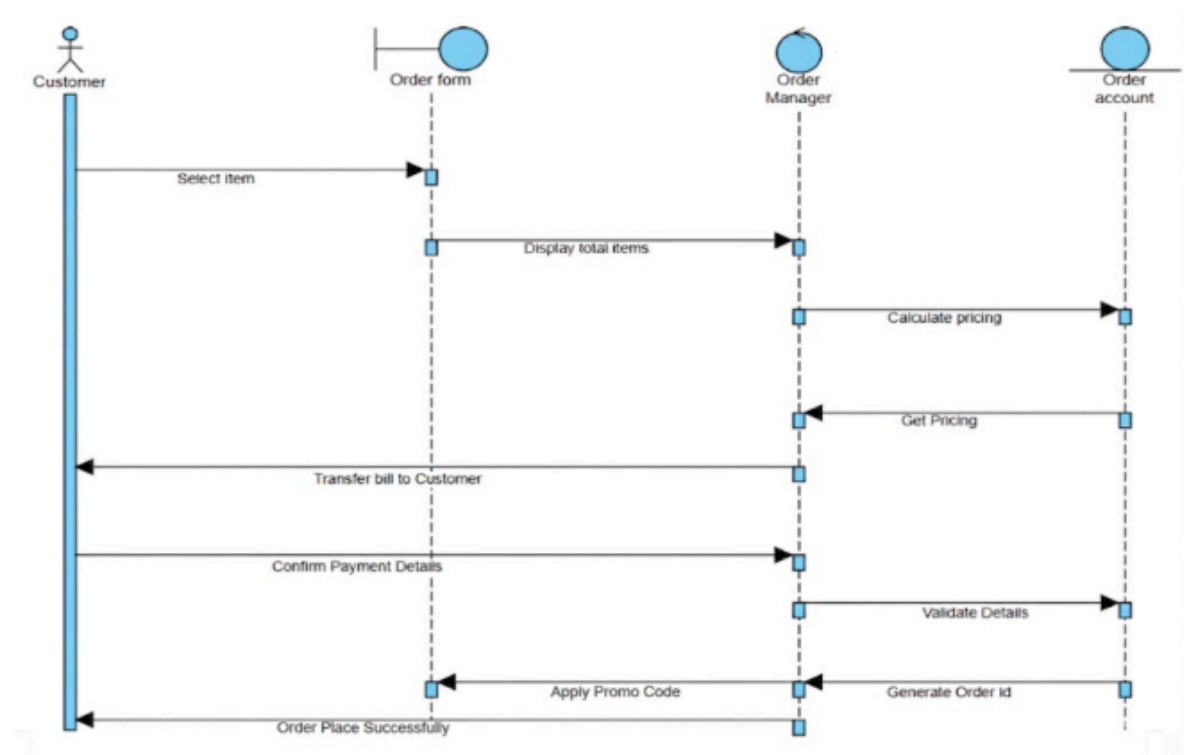
Updates the specific order record and records the completion time for the rider's performance log.

Sequence Diagram:

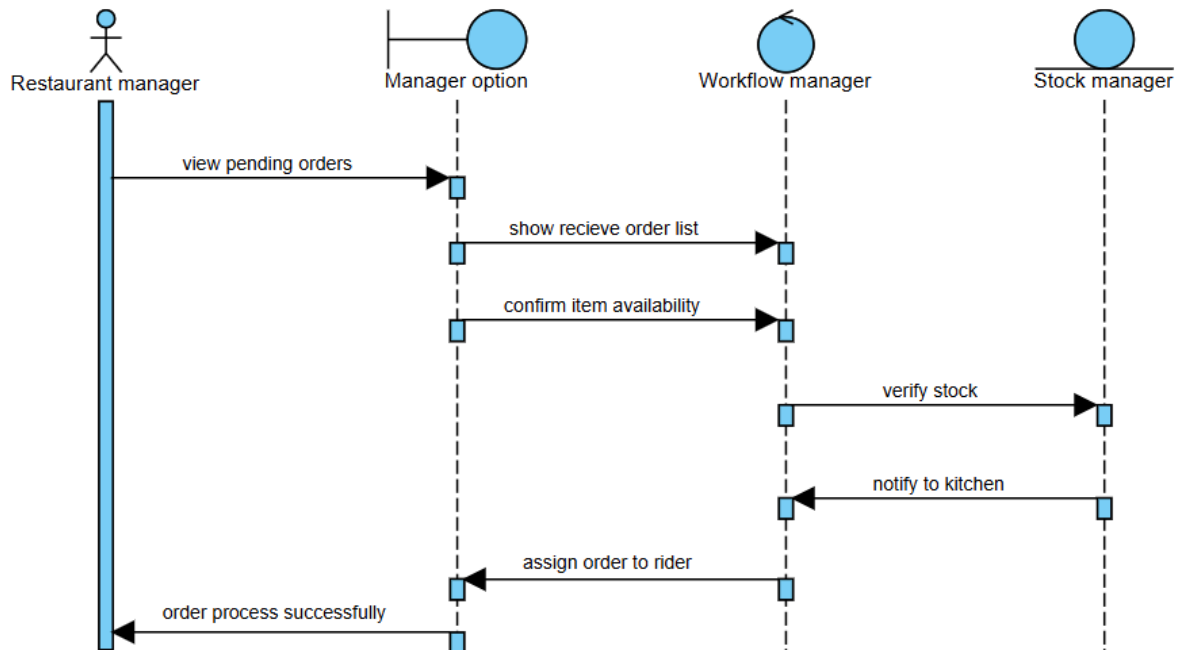
a) Login/Signup:



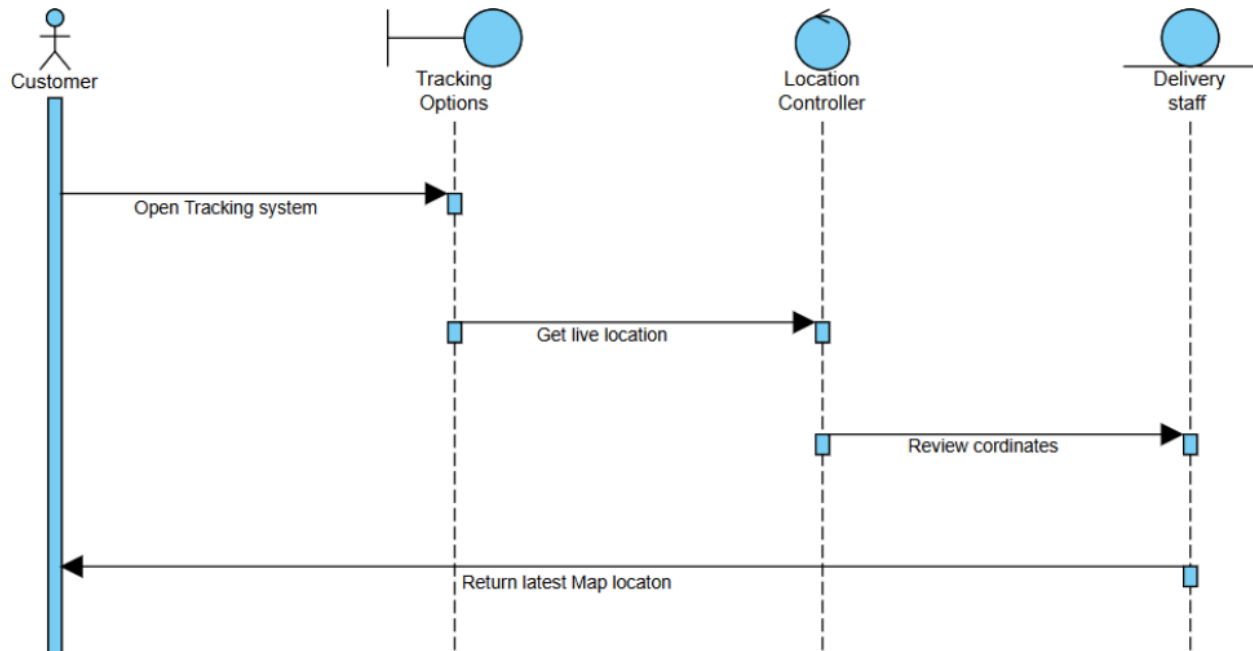
b) Place Order



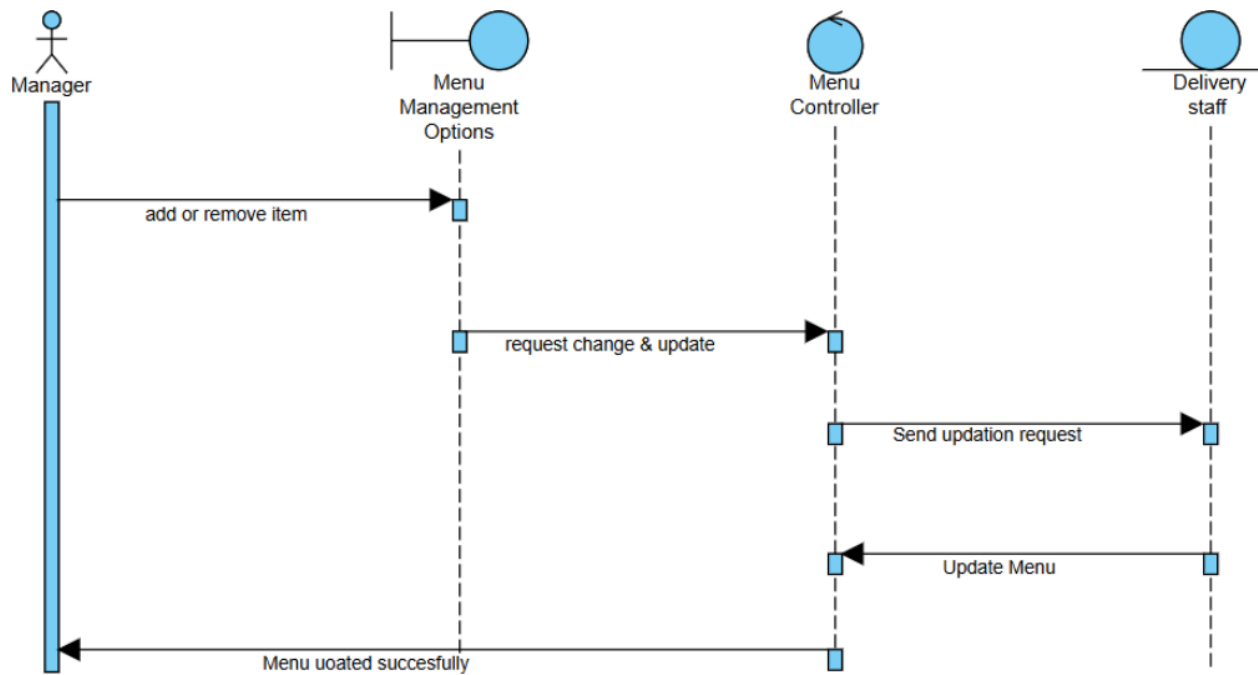
c) Process order



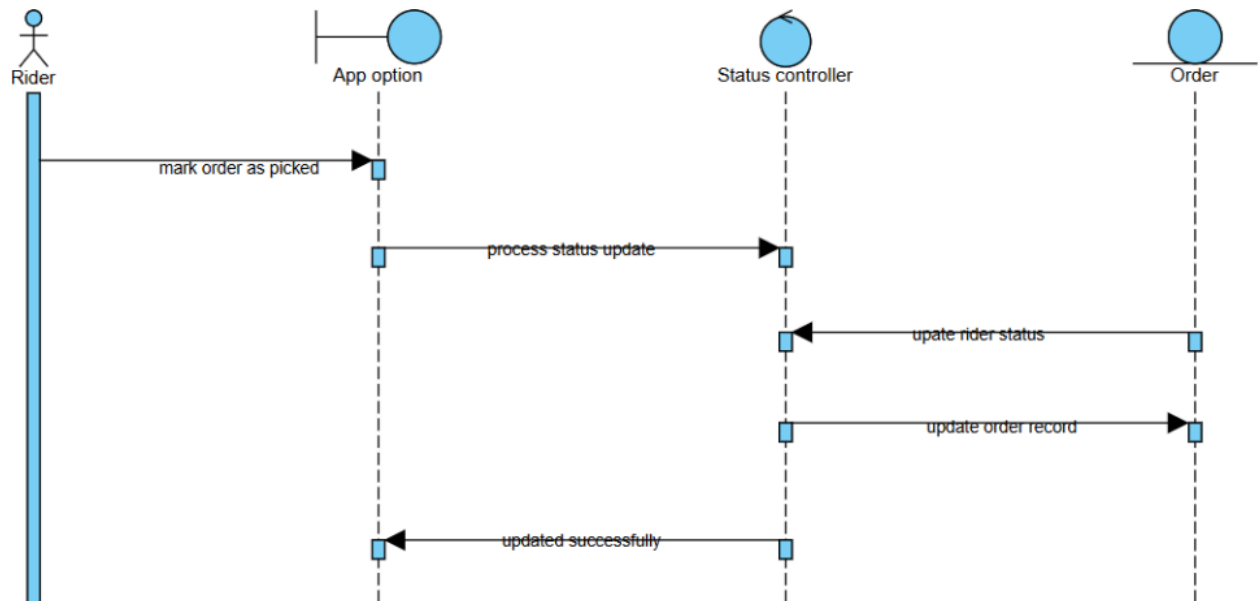
d) Track Order



e) Manage Menu



f) Update Delivery Status



2.4 Operating Environment

The Online Food Delivery Management System is designed to operate seamlessly across different platforms to accommodate all user classes. The software will coexist and function within the following environment:

Users:

Customers & Managers: Can access the system via standard web browsers such as Google Chrome running on Windows.

Delivery Riders & Customers: Can also access the system via a native mobile application environment.

Backend & Database Server: The backend system logic and database will run on a cloud-based server environment such as Linux Server to securely manage user authentication data, menu structures, and live order logs.

2.5 Design and Implementation Constraints

Some limitations and constraints must be considered during development such as:

- The platform must work smoothly on both mobile devices and computers.
- The system should be able to handle at least 500 users at the same time.
- The user interface should remain simple and easy to use so that customers can use this system without any difficulty.
- The development timeline for the project is expected to be between 2 and 3 months.

2.6 User Documentation

The following documentation will be provided to ensure all users can operate the system smoothly:

- A comprehensive PDF guide covering step-by-step instructions for Customers, Managers, and Riders.
- A dedicated section within the mobile app to assist users with common issues and order tracking.
- Detailed documentation for System Administrators regarding database management and security settings.
- On-screen prompts for first-time users to help them navigate the interface and place their first order.

2.7 Assumptions and Dependencies

The successful operation of the Food Delivery Management System assumes that all users have access to a stable internet connection (3G/4G/5G or Wi-Fi) and that their devices meet the minimum hardware requirements. A critical dependency is the continuous availability and accuracy of the Google Maps for real-time location tracking and navigation. It is also assumed that the third-party restaurants will provide timely updates to their menus and that delivery

personnel will have GPS-enabled smartphones. If any of these external services face downtime or if users lack connectivity, the system's core functionalities, such as order tracking and automated dispatching, will be significantly affected.

3. External Interface Requirements

3.1 User Interfaces

The system requires three distinct user interfaces. All screens will follow a clean, consistent layout with standard error messages displayed at the bottom of the screen.

Customer Interface (Web & Mobile):

A grid-style layout for the *Browse Menu* screen with clear food pictures and prices.

An interactive tracking screen with an embedded map widget for the *View live map* feature.

Standard buttons like Place Order, Apply Promo, and Submit Review will be easily clickable.

Restaurant Manager Dashboard :

A split-screen layout designed for desktop views.

One side shows list controls to *Manage Menu* buttons to add, remove, or *Update Prices*.

The other side displays an active list pane to *Process order* with quick actions to accept, *Reject Order*, or *Assign Order* to riders.

Delivery Rider App:

A dark-mode friendly, high-contrast mobile layout for outdoor use.

Features large list cards displaying customer details under *View Location*.

Includes prominent, easy-to-tap toggle buttons to *Update Status* such as Marked picked up and Mark delivered while driving.

3.2 Hardware Interfaces

No Hardware is required for this but this platform is compatible with devices such as:

- Computers
- Laptops
- Smartphones

- Tablets

3.3 Software Interfaces

The application relies on specific software interfaces to process and synchronize data between the front-end clients and the back-end system:

Operating Systems: The software interfaces with Android and iOS for mobile applications, and handles standard web browser engines for desktop management.

Google Maps: This is the main external software link. The system sends location coordinates to this API and receives mapped routes and live visual tracking data to power the *View live map* and *Use Google Maps* use cases.

Database Management Interface: The application logic interacts with a central database SQL database server using standard queries.

3.4 Communications Interfaces

The system uses HTTPS as the primary protocol to ensure all data transferred between the mobile app and server is encrypted and secure. Real-time updates and order alerts are handled through Firebase Cloud Messaging, while JSON is used as the standard format for fast and lightweight data exchange. Additionally, the system employs SMTP for sending automated emails like receipts and account verifications, ensuring all users stay synchronized with the latest order status.

4. System Features

4.1 System Feature 1 (User Authentication and Profile Management)

4.1.1 Description and Priority

This feature allows users to create accounts and securely access the application. It maps directly to the Signup, Login, and View details use cases.

- **Priority:** High

4.1.2 Stimulus/Response Sequences

- **Stimulus:** User enters credentials on the login/signup screen and clicks submit.
- **Response:** The system validates the inputs, checks the database, grants access to the correct dashboard (Customer, Manager, or Rider), and displays their profile details.

4.1.3 Functional Requirements

REQ-AUTH-1: The system shall allow new users to register an account by providing mandatory details like name, email, phone number, and password (*Signup*).

REQ-AUTH-2: The system shall verify credentials against the database when a user attempts to access their account (*Login*).

REQ-AUTH-3: The system shall fetch and display the logged-in user's profile and account settings from the database automatically (*View details*).

REQ-AUTH-4: If the entered credentials do not match any record, the system shall show a simple "Invalid Email or Password" error message.

4.2 System Feature 2 (Customer Order Lifecycle and Tracking)

4.2.1 Description and Priority

This feature handles everything a customer does, from looking at food items to placing an order, paying, and tracking it on a map. This maps to *Browse Menu*, *Place Order*, *Apply promo code*, *Make payment*, *Track Order (View live map)*, and *Rate or Review*.

- **Priority:** High

4.2.2 Stimulus/Response Sequences

- **Stimulus:** Customer selects food items from the menu, inputs a promo code, and clicks checkout.
- **Response:** The system calculates the total amount, processes the payment action, generates an active order tracking link, and lets the user track it on a live map.

4.2.3 Functional Requirements

REQ-ORD-1: The system shall display an active list of available food items with updated prices for the customer (*Browse Menu*).

REQ-ORD-2: The system shall allow the customer to select items and create a new delivery order request (*Place Order*).

REQ-ORD-3: The system shall allow customers to apply a valid discount code to deduct the total order bill (*Apply promo code*).

REQ-ORD-4: The system shall process the transaction amount before finalizing the order request (*Make payment*).

REQ-ORD-5: The system shall immediately cancel the checkout flow and show a "Payment Failed" error if the customer's payment transaction is unsuccessful (*Reject Order*).

REQ-ORD-6: The system shall show a real-time status tracker dashboard to the customer after a successful order placement (*Track Order*).

REQ-ORD-7: The system shall embed a live Google Maps view on the tracking screen to show the delivery rider's moving coordinates (*View live map*).

REQ-ORD-8: The system shall allow the customer to input a 1-5 star rating and type a text comment after the order status is marked as delivered (*Rate or Review*).

4.3 System Feature 3 (Restaurant and Rider Operations)

4.3.1 Description and Priority

This feature covers the backend actions taken by the manager and the rider to prepare and deliver the food. It maps to *Manage Menu* (*add or remove item, Update Prices*), *Process order* (*Confirm availability, Reject Order*), *Assign Order, Update Status* (*Marked picked up, Mark delivered*), and *View Location* (*Use Google Maps*).

- **Priority:** High (Handles operational fulfillment).

4.3.2 Stimulus/Response Sequences

- **Stimulus:** Manager receives a new order request or changes a menu item price. Rider receives an assigned order and updates checkpoints.
- **Response:** The system updates the menu list, syncs order statuses across all dashboards, and routes navigation paths using external map data.

4.3.3 Functional Requirements

- **REQ-OPS-1:** The system shall allow the manager to add new dishes or delete old items from the visible restaurant digital menu (*add or remove item*).
- **REQ-OPS-2:** The system shall allow the manager to alter the pricing values of any item on the menu instantly (*Update Prices*).
- **REQ-OPS-3:** The system shall display incoming pending orders on the manager's dashboard for operational review (*Process order*).
- **REQ-OPS-4:** The system shall require the manager to check and verify stock levels before accepting a incoming customer request (*Confirm availability*).
- **REQ-OPS-5:** The system shall allow the manager to cancel an order manually if items are out of stock or kitchen capacity is full (*Reject Order*).
- **REQ-OPS-6:** The system shall allow the manager to select a delivery worker from a list of active personnel and link them to a confirmed order (*Assign Order*).
- **REQ-OPS-7:** The system shall show the customer's delivery destination address and phone number on the assigned rider's mobile view (*View Location*).
- **REQ-OPS-8:** The system shall trigger the external Google Maps application with pre-loaded coordinates when the rider requests navigation help (*Use Google Maps*).
- **REQ-OPS-9:** The system shall provide simple toggle options for the rider to change the order status landmark flags (*Marked picked up* and *Mark delivered*).

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Response Time: The system must load the *Browse Menu* list within 2 seconds of a customer request under normal network conditions.

Real-time Tracking Updates: The *View live map* and *Use Google Maps* features must sync location coordinates every 5 seconds to keep data fresh.

Concurrent Users: The backend database must handle at least 500 active orders and user interactions at the same time without slowing down.

5.2 Safety Requirements

Data Loss Prevention: The system must auto-save active order details instantly in the database, so if a user's phone turns off or crashes during *Place Order*, their cart/order progress isn't lost.

Navigation Safety: The mobile app interface for the Delivery Rider must have a simple, clean UI with minimal text to avoid causing high visual distractions while driving.

5.3 Security Requirements

Secure Authentication: User passwords entered during *Signup* and *Login* must be encrypted using strong hashing techniques before saving them to the database.

Access Control: A user cannot access functionalities belonging to other roles.

Payment Security: All data transmitted during the *Make payment* use case must use secure HTTPS protocols to protect user transaction logs.

5.4 Software Quality Attributes

Availability: The core system must fully maintain uptime so customers can place orders at any time during business hours.

Usability: The application UI must be easy to navigate, meaning a new customer should be able to complete a *Place Order* workflow within 3 simple screens.

Reliability: If the payment connection fails, the system must reliably trigger the *Reject Order* exception immediately to prevent double-charging the customer.

5.5 Business Rules

User Authority: Only the authorized Restaurant Manager account can perform the *Update Prices* and *add or remove item* tasks.

Order Rejection Limits: The system allows the manager to execute *Reject Order* only *before* the rider status is changed to *Marked picked up*.

Promo Code Validity: A customer can successfully apply a coupon code through *Apply promo code* only if it matches active date ranges and minimum order price requirements set by the system logic.

6. Other Requirements

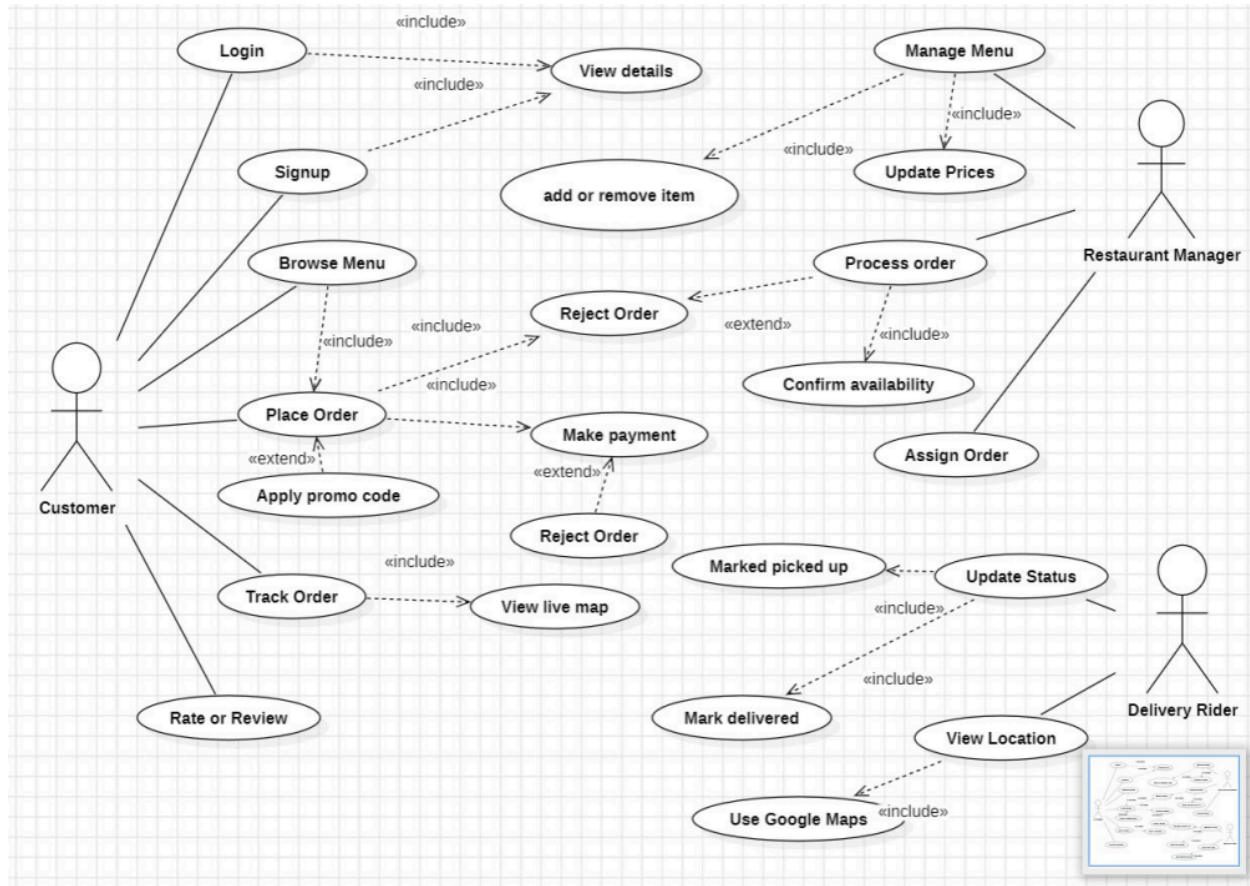
No other requirements

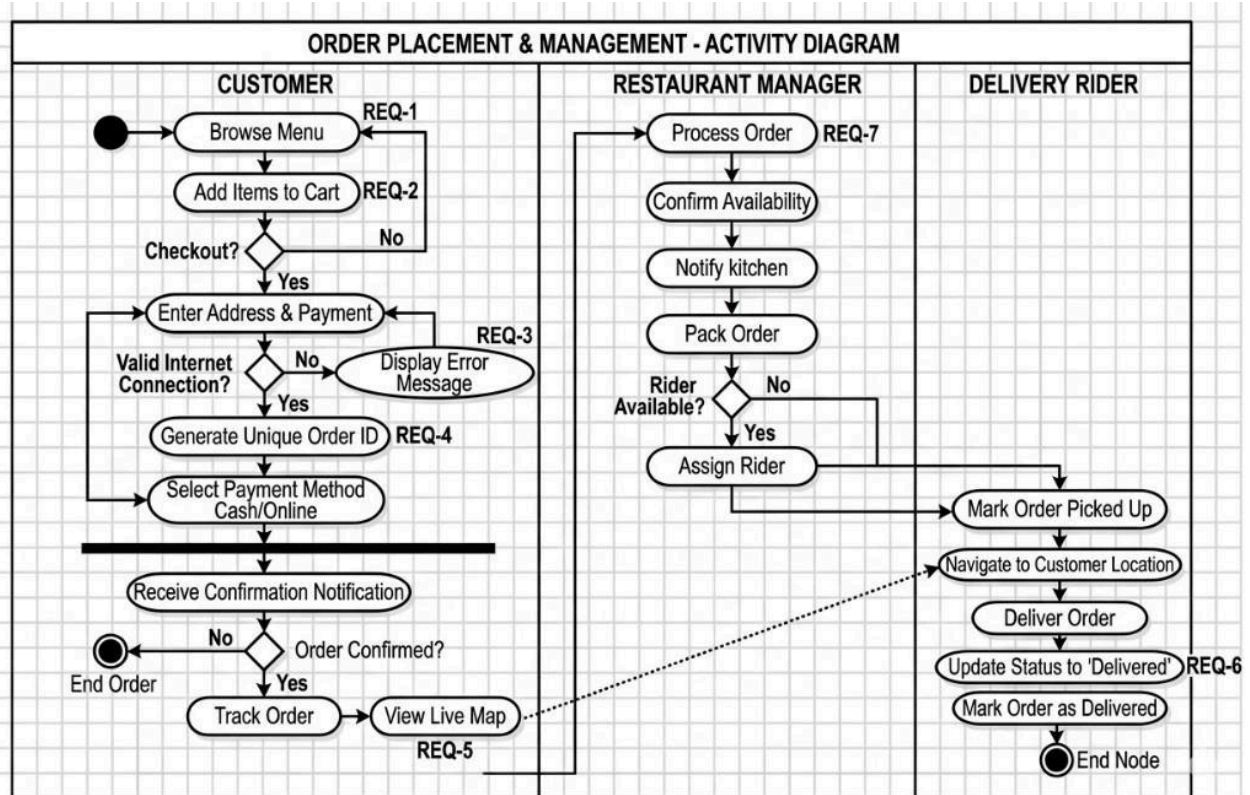
Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

Use Case Diagram:

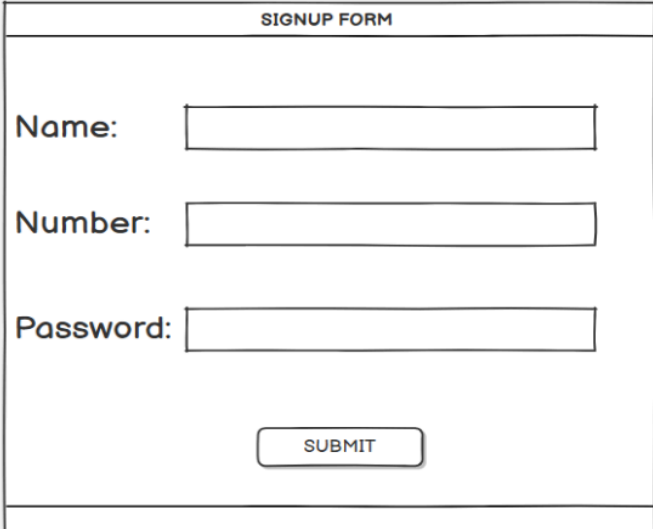


Activity Diagram:**Appendix C: To Be Determined List**

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>

Prototypes:

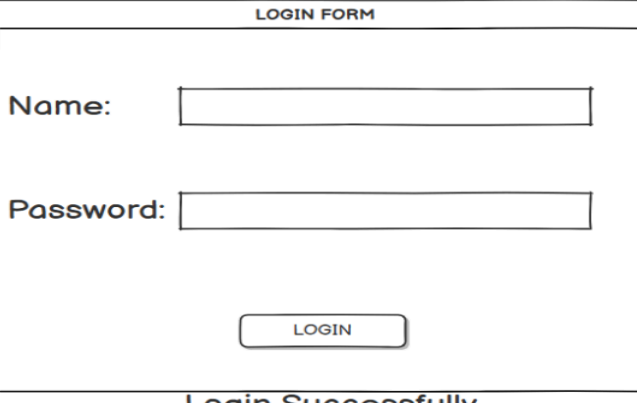
Signup Form:



A wireframe of a signup form. It has a title bar labeled "SIGNUP FORM". Below the title bar, there are three input fields: "Name:", "Number:", and "Password:". Each field is represented by a rectangular box. Below the input fields, there is a rounded rectangular button labeled "SUBMIT".

- After signing up, customer can login.

Login page:



A wireframe of a login form. It has a title bar labeled "LOGIN FORM". Below the title bar, there are two input fields: "Name:" and "Password:". Each field is represented by a rectangular box. Below the input fields, there is a rounded rectangular button labeled "LOGIN". At the bottom of the form, there is a status bar labeled "Login Successfully".

- After login, customer enters in place order section.

Menu page:

A screenshot of a mobile application screen titled "MENU". The screen displays four food categories, each with a corresponding checkbox: "BBQ:", "FAST FOOD:", "CHINESE FOOD:", and "BEVERAGES:". Below these categories is a rounded rectangular button labeled "CHECKOUT". The screen has a simple border and a header bar.

- Then customer enters into payment process.

Payment Option:

A screenshot of a mobile application screen titled "PAYMENT OPTION". The screen displays three payment methods, each with a corresponding checkbox: "C.O.D:", "Visa Card:", and "Online Transfer:". Below these options is a text input field labeled "Apply Promo Code". At the bottom of the screen, there are two buttons: "APPLY" and "DONE". The screen has a simple border and a header bar.

Payment Details:

PAYMENT DETAILS	
Name:	
Account Type:	
Card Number:	
Your total amount is : 3000	
ORDER	
Your order is placed successfully	

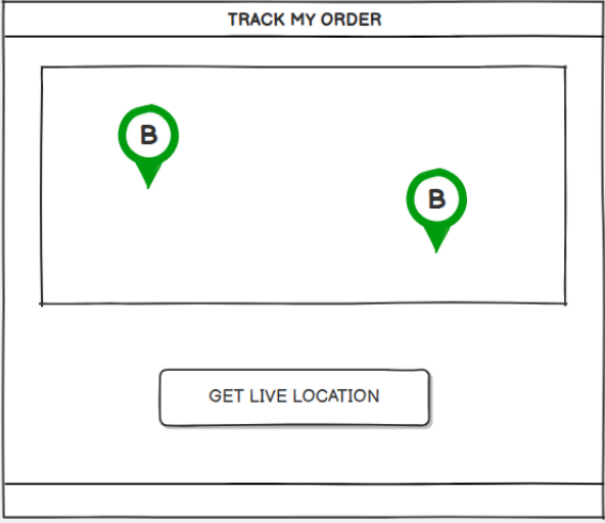
MANAGER'S SCREEN

Received Orders:

RECIEVED ORDERS
RECIEVED ORDERS a) BBQ b) Chinese c) Fast Food d) BBQ e) Fast Food
☐ Items are available in stock ☐ Items are not available in stock
CONTINUE
ORDER PROCESS SUCCESSFULLY

CUSTOMER'S SCREEN

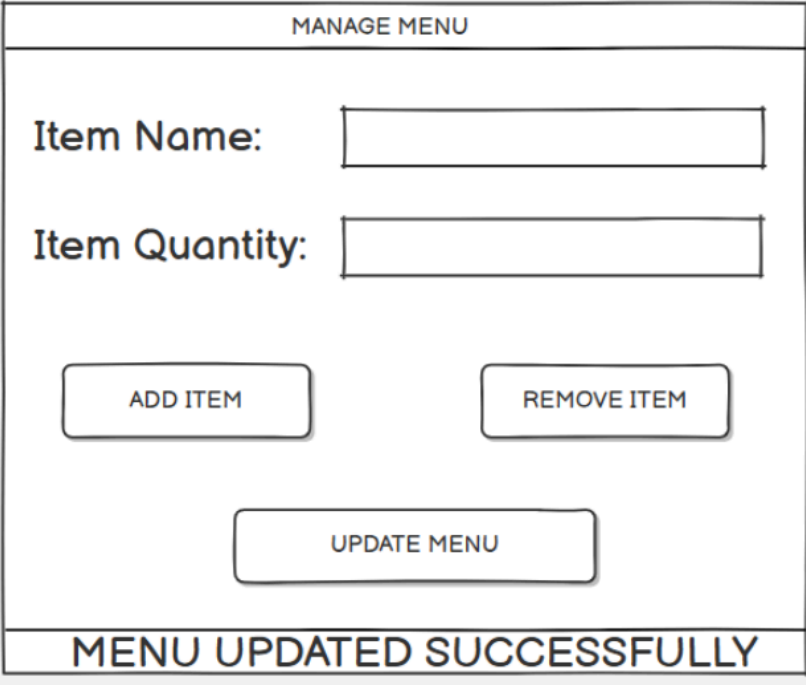
Track My Order:



The 'TRACK MY ORDER' screen features a header with the title 'TRACK MY ORDER'. Below the header is a large rectangular map area containing two green location pins, each with a white circle and the letter 'B' inside. At the bottom of the screen is a button labeled 'GET LIVE LOCATION'.

MANAGER'S SCREEN

Manage Menu:



The 'MANAGE MENU' screen has a header with the title 'MANAGE MENU'. Below the header are two input fields: 'Item Name:' followed by a text box, and 'Item Quantity:' followed by a text box. Under these fields are three buttons: 'ADD ITEM' and 'REMOVE ITEM' are side-by-side, and 'UPDATE MENU' is centered below them. At the bottom of the screen is a status bar that reads 'MENU UPDATED SUCCESSFULLY'.

Update Delivery Status:

A screenshot of a web form titled "UPDATE DELIVERY STATUS". The form has a header bar with the title. Below the header, there are two radio button options: "Mark order as picked" and "Not picked yet". Below these options is a button labeled "UPDATE DELIVERY STATUS". At the bottom of the form, there is a status bar that says "UPDATED SUCCESSFULLY".

CUSTOMER'S SCREEN

Rate & Review:

A screenshot of a web form titled "RATE AND REVIEW". The form has a header bar with the title. Below the header, there is a section for rating with the text "GIVE YOUR RATING:" followed by five empty star icons. Below the rating section, there is a section for writing a review with the text "WRITE YOUR REVIEW:". Below this text is a large text area with the placeholder text "YOUR FEEDBACK IS IMPORTANT...". At the bottom of the form, there is a button labeled "SUBMIT".

Class Diagram:

